More Than the Sum of Its Parts? Markups and the Role of Establishments*

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Abstract

We study misallocation and optimal policy under size-dependent establishment-level market power. Using administrative data on Sweden's services industries, we provide evidence that market power increases with establishment size rather than firm size, and document that multi-establishment firms are rare but large and open successively smaller establishments. We develop a macroeconomic model guided by these findings. We characterize distortions to firms' choices of which establishments to open and how much to produce at each. Firms overvalue small relative to large establishments along both margins, and overvalue opening new establishments relative to expanding existing ones. In the calibrated model, optimal firm size-dependent policy subsidizes large single-establishment firms relative to virtually all larger, higher markup multi-establishment firms, reversing the prescription from models with firm-level market power. This reflects a tension between the benefits of undoing excessive establishment opening and expanding high markup establishments.

Keywords: Multi-establishment firms, Market power, Misallocation, Size-dependent markups, Size-dependent policy

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1 Introduction

A longstanding concern with large firms is that they have market power: they face a low price elasticity of demand, so they restrict production to keep prices high. This implies a misallocation of inputs away from larger firms with higher markups of price over marginal cost, toward smaller firms with lower markups. The optimal tax schedule is therefore decreasing in firm size in order to undo this distortion. More generally, changes favoring large firms, as in the recent rise of industry concentration in the U.S., may reflect an improvement in allocative efficiency rather than a loss.[1] Evidence that markups are increasing in firm size lends support to these theories.[2]

This reasoning treats firm output as one-dimensional. However, a growing literature documents the importance of multi-unit firms that expand both by selling more through existing units, such as establishments, products, or customer relationships, and by adding new units. This distinction between intensive and extensive margins matters when market power is determined at the unit level rather than the firm level. For example, a coffee chain may have market power at a particular establishment that dominates local competitors, regardless of the success of its other establishments. Whereas firm-level market power discourages production along all margins equally, unit-level market power implies potentially different distortions across firms' various intensive and extensive margin decisions. What are these distortions? Is it still efficient to shift production toward large, high markup firms?

We answer these questions using a macroeconomic model with size-dependent establishment-level market power, informed by administrative data on Swedish services firms and their establishments. We show that firm markups are a misleading guide to optimal reallocation and policy. In the calibrated model, optimal firm size-dependent policy subsidizes large single-establishment firms relative to virtually all larger, higher markup multi-establishment firms, reversing the prescription from models with firm-level market power. Our theoretical characterization of distortions explains why: firms overvalue opening new establishments relative to expanding existing ones of the same size. This bias is even stronger when their marginal establishments are relatively small, as our empirical evidence suggests. Thus, high firm markups call for expanding existing establishments, but not necessarily opening new ones.

[1] See Baqaee and Farhi (2020) for a general analysis and Weiss (2020) for a particular theory of the rise in concentration where it is associated with an increase in welfare.

[2] See Edmond et al. (2023) for evidence in the U.S., and Burstein et al. (2025) for France.

Our first contribution is empirical. Using administrative data on the universe of Swedish services firms and establishments (physical locations) from 1997 to 2017, we document three findings. First, multi-establishment firms are rare but large, making up 2.7% of firms but earning 49% of sales because they have both many and large establishments compared to industry competitors. Second, within industries, firms' markups are positively related to their sales per establishment but not their number of establishments once we control for both. This is consistent with market power increasing in establishment size rather than firm size. Third, successive establishments within firms have lower employment and wage bills, even controlling for establishment age, firm age, year, and municipality-industry. We interpret this as firms working down a ranked set of establishment opportunities, so their marginal establishments are smaller, with less market power, than their average establishments.

Informed by these findings, we develop a macroeconomic model where each firm has an initial establishment and a set of additional establishment opportunities that differ by quality, which shifts demand, and opening cost. Firms use a fixed aggregate supply of labor to open establishments and produce at them. Demand follows a non-CES Kimball (1995) aggregator over establishments, where larger establishments face a lower price elasticity of demand, meaning they have more market power. As such, a firm's markup—its cost-weighted average establishment markup—is increasing in the size of its establishments, not its number of establishments, as in the data.

Our second contribution characterizes the resulting distortions. If demand becomes strictly less elastic with establishment size, then we can raise welfare by 1) shifting production from smaller, lower markup establishments to larger, higher markup ones, 2) closing smaller marginal establishments at one firm and using the labor to open larger ones at another multi-establishment firm, and 3) closing marginal establishments to free up labor for expanding production at weakly larger establishments, or even somewhat smaller ones. The last comparison between firms' intensive and extensive margins is especially powerful if marginal establishments are relatively small, as suggested by our empirical evidence.

Intuitively, we can raise welfare by shifting labor toward activities undervalued by private incentives. For expanding an establishment's output, marginal revenue is less than the social value, the price, because selling an additional unit requires cutting the price on inframarginal units. Larger establishments face less elastic demand, so they must cut their price more to increase production, which means they set higher

markups and produce too little. For opening a new establishment, what matters is *total* establishment revenue as a share of the social value of *total* establishment output. This share also falls with establishment size, but less steeply than the share captured on the marginal unit. Firms therefore undervalue opening larger establishments relative to smaller ones, and expanding production at existing establishments relative to opening new ones. By contrast, with a constant demand elasticity, revenue captures the same share of social value across all activities, so there is no misallocation.

Although we focus on potentially multi-establishment firms with establishment-level market power, this theoretical analysis is general and can also apply to multi-product or multi-customer firms with product- or customer-level market power.

Our third contribution uses the Swedish services data to quantify distortions and draw out the implications for misallocation and policy. We calibrate the model to match multi-establishment firms' sales share, frequency, and the decomposition of their sales into many vs. large establishments. We also match the decline in size across successive establishments within firms, the relationship between firm markups and sales per establishment, and the cost-weighted average markup. In the market equilibrium, sufficiently large firms are multi-establishment. Larger firms also have both more and larger establishments, with the latter driving higher markups.

We confirm two key model predictions in the data. First, differences in sales among larger firms increasingly reflect differences in their number of establishments rather than in their average establishment size. Second, for this reason, whereas the relationship between log firm markups and log sales per establishment steepens in the upper tail, the relationship with log firm sales is flatter and nearly linear.

We then solve for the first best allocation, which raises aggregate output by 1.52% relative to the market equilibrium. The resulting changes in firm output are non-monotonic in firm size and markups: the largest single-establishment firms expand, but 93% of the even larger, higher markup multi-establishment firms contract because establishment closures outweigh increases in output per establishment. This reflects a tension between their high markups, which call for more production at existing establishments, and their overvaluation of opening new establishments. Thus, a high markup does not imply that a firm should expand overall.

Turning to policy, we study the optimal tax schedule conditioning only on firm sales, motivated by real-world firm size-dependent taxes and regulations and by the difficulty of conditioning on sales at individual establishments or, in other contexts,

products or customers. Holding fixed the set of open establishments, optimal policy imposes lower tax rates on larger firms to undo the distortions implied by their higher markups. Aggregate output rises by 0.81%, achieving almost all the potential gains of 0.85% from reallocation in this setting. But implementing this policy backfires if the set of establishments responds in equilibrium: output *falls* by 2.32% because large firms respond to their relative subsidies by opening too many establishments.

Accounting for firms' full responses, optimal firm size-dependent policy becomes non-monotonic. Larger firms' higher markups call for lower tax rates, but the need to restrain their establishment opening pushes in the opposite direction. Consequently, large single-establishment firms face lower tax rates than more than 99.99% of larger, higher markup multi-establishment firms.

Firm size-dependent policy is also less effective when the set of open establishments can adjust, despite greater potential gains from reallocation. That is, optimal policy raises aggregate output by only 0.60%, compared with 0.81% when the set of establishments is fixed. This reflects the inability of policy that conditions only on total firm sales to separately incentivize the intensive and extensive margins. Optimally designing firm size-dependent policy jointly with a flat tax on opening establishments resolves this tension. The tax schedule is again monotonic, giving lower rates to larger firms, and output rises by 1.48%, recovering nearly all the first best gains.

Finally, our theoretical and quantitative results are robust to uniform pricing across establishments within firms. We also extend our characterization of distortions to incorporate heterogeneous establishment productivity, industries, and locations. It is optimal to shift production toward locations dominated by few establishments, as in Becker et al. (2026). This affects firm size-dependent policy if a firm's size is strongly related to the level of concentration in the locations where it operates.

Related literature. We contribute to a growing literature studying multi-unit firms that expand by selling more through existing units and by adding new ones. Cao et al. (2022) and Hsieh and Rossi-Hansberg (2023) document the importance of expansion through new establishments for the recent rise of large firms in U.S. services. Oberfield et al. (2024) develop a model of firms' establishment location choices under constant market power and markups. Related work studies the role of firms acquiring new customers (Arkolakis, 2010; Einav et al., 2022) and introducing new products (Bernard et al., 2011; Argente et al., 2024).

Two papers in this vein are especially close. Mayer et al. (2014) develop a model with size-dependent product-level markups to study how competition affects sales across products, including which have positive sales. A key difference is that firms in our model weigh establishments' operating profits against opening costs, so marginal establishments have positive sales and market power. Afrouzi et al. (2025) study concentration and misallocation in a model with size-dependent customer-level market power and endogenous customer acquisition. They document that firms' markups are increasing in their sales per customer but not their number of customers when controlling for both, analogous to our empirical result on markups. Two contrasts are that their model has homogeneous customers within firms and no lower bound on the size of firms' customer bases, whereas we incorporate heterogeneous establishments within firms and a single-establishment lower bound, both informed by our data.

We add to this literature by characterizing how size-dependent establishment-level market power distorts firms' decisions of which establishments to open and how much to produce at each. We also differ from previous work by drawing out the implications for firm size-dependent policy. Our analysis is guided by novel evidence on how firm markups relate to establishment vs. firm size, how establishment size varies within firms, and the prevalence of small firms at the single-establishment lower bound.

Our policy analysis builds on Edmond et al. (2023), who study a model where market power and markups are increasing in *firm* size. They compute the implied costs and show that optimal policy gives lower tax rates to larger, higher markup firms. We demonstrate that this prescription need not hold under establishment-level market power because of a conflict between the distortions to firms' intensive vs. extensive margins. Becker et al. (2026) allow a firm's market power to vary across local markets, similar to market power varying across establishments within firms in our model. However, they take firms' participation in local markets as given, abstracting from the extensive margin decisions central to our results.

Finally, our analysis of intensive vs. extensive margin distortions builds on a long line of theoretical work on distortions to production and entry under monopolistic competition with size-dependent market power, including Dixit and Stiglitz (1977), Zhelobodko et al. (2012), Dhingra and Morrow (2019), Behrens et al. (2020), and Matsuyama and Ushchev (2021). We contribute by adding a *firm-specific* entry choice, with each firm choosing which of heterogeneous establishment opportunities to open. Market power depends on size at the establishment level rather than the firm level,

so distortions to production and entry decisions vary both within and across firms.

The paper proceeds as follows. Section 2 contains our empirical analysis. In Section 3, we describe the model and prove our theoretical result, which characterizes the distortions implied by establishment-level market power. We calibrate the model in Section 4, which we use to study misallocation and optimal firm size-dependent policy in Section 5. In Section 6, we extend our theoretical result to incorporate uniform pricing and richer establishment-level heterogeneity. Section 7 concludes.

2 Empirical Analysis

We analyze data on the universe of Swedish firms and establishments in services industries from 1997 to 2017. We first examine the role of the intensive margin (establishment size) and extensive margin (set of open establishments) for firm sales. We then estimate how each margin relates to firm markups. Finally, we study the size of firms' successive establishments to learn about their marginal establishments.

Data. We merge two administrative datasets from Statistics Sweden, the government agency responsible for official statistics. Företagens Ekonomi provides all firms' annual sales, expenditures on intermediate inputs, and physical capital stocks. Registerbaserad arbetsmarknadsstatistik provides all monthly employer-employee linkages, with workers' earnings and their firm and establishment (physical location) identifiers. Aggregating workers within establishment-years yields annual data on each firm's set of establishments and each establishment's wage bill, employment, municipality,[3] and industry. We deflate nominal variables to 2017 SEK using the GDP deflator.

We restrict attention to services: for our firm- and establishment-level analyses, respectively, we include firms and establishments with positive sales and employment operating in the private economy in 5-digit services industries.[4] This leaves about 3.5 million firm-year observations, covering about 60% of sales in the Swedish economy.

[3]There are 291 municipalities in 2017, ranging from Stockholm with about 1 million services workers to Bjurholm with only 960.

[4]5-digit is the most disaggregated level. Of 821 industries, we keep codes 45 and above, excluding "financial and insurance activities". This leaves 423 services industries with an average of 422 firms.

2.1 Intensive vs. Extensive Margins of Sales

A firm's sales are the product of its sales per establishment (the intensive margin) and number of establishments (the extensive margin). The vast majority of firms are single-establishment so only the intensive margin is relevant. But both margins are important for the largest firms, which tend to be multi-establishment.

Table 1 shows that multi-establishment firms are only 2.7% of services firms but earn 48.7% of sales. This reflects both more and larger establishments: they have 6.2 times as many establishments as the average firm and their establishments are 2.9 times as large as the average establishment. Moreover, multi-establishment firms with more establishments tend to have larger ones. Specifically, regressing log sales per establishment on log number of establishments among multi-establishment firm-years, controlling for industry-year fixed effects, yields an estimated elasticity of 0.143.[5]

Table 1: Multi-Establishment Firms vs. All Firms

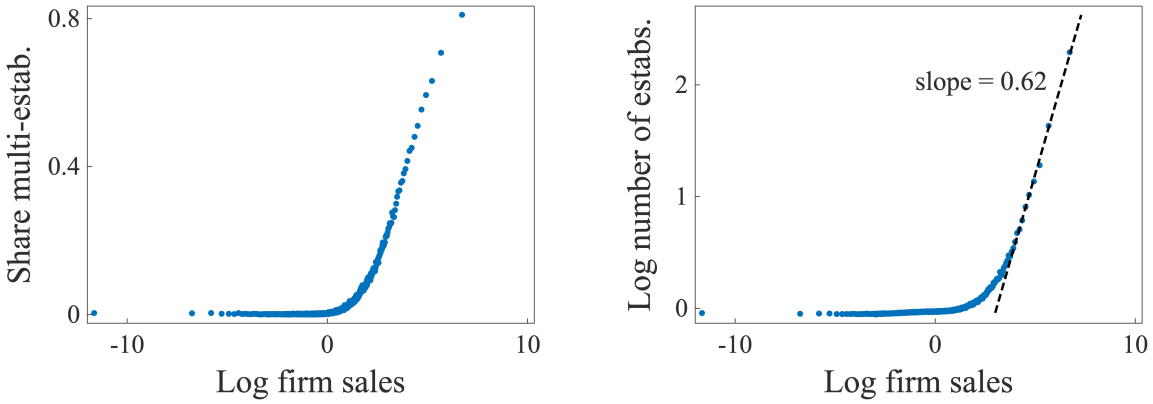
Multi-establishment firms		**Decomposing multi-establishment firm sales**	
Firm share	*Sales share*	*Relative number of establishments per firm*	*Relative sales per establishment*
2.7%	48.7%	6.2	2.9

Statistics computed for services industries in each year and averaged arithmetically across years. Column 3 divides the average number of establishments for multi-establishment firms by the average for all firms; column 4 divides the average size of multi-establishment firms' establishments by the average for all establishments.

Figure 1 provides a more detailed comparison across firms based on their sales relative to industry-year means. For most of the relative sales distribution, firms are overwhelmingly single-establishment (left panel), so sales differences are driven almost entirely by sales per establishment rather than number of establishments (right panel). But among the firms with the highest relative sales, the majority are multi-establishment and relative number of establishments plays an increasing role, explaining 62% of the difference in log relative sales between the two highest bins.

[5]The robust firm-clustered standard error is 0.014, with an R^2 of 0.374, using 93,433 observations.

Figure 1: The Role of the Extensive Margin



Firm-years grouped into 1000 equally sized bins, ranked by sales relative to industry-year means. Each dot plots the share that are multi-establishment (left) and average log number of establishments (right, relative to industry-year means) within a bin against average log relative sales. The dashed line shows the slope between the top two bins.

2.2 Size-Dependent Markups

We now investigate the relationship within industries between firms' markups and the intensive and extensive margins of their sales, i.e., sales per establishment and number of establishments. We focus on firm markups because we do not observe establishment sales so we cannot estimate establishment markups. We find that firm markups are positively associated with sales per establishment but not number of establishments once we control for both. This is evidence that markups, and therefore market power, are increasing in size at the establishment level rather than the firm level.

To proxy for a firm's markup relative to industry-year competitors, we use its ratio of nominal sales to expenditures on intermediate inputs. If intermediates are chosen flexibly at a linear price, the markup is this ratio times the elasticity of output with respect to intermediates. Assuming a constant elasticity across firms within industry-years, differences in log markups equal differences in log sales-to-intermediates ratios.[6] This firm markup is the intermediates-weighted average of establishment markups.

We regress our markup proxy on firm sales and its intensive and extensive margins:

$$\ln(\text{sales}/\text{intermediates})_{i,j,t} = \beta \ln(\text{size})_{i,j,t} + \text{fixed effects}_{j,t} + \epsilon_{i,j,t}. \tag{1}$$

[6] De Loecker and Warzynski (2012) and De Ridder et al. (2026) discuss this estimation approach.

An observation is a firm-year pair, indexed by firm i, industry j, and year t. The coefficient of interest, β, can be on any subset of log sales, log sales per establishment, and log number of establishments. Industry-year fixed effects absorb industry-year-specific output elasticities, so β is the within industry-year elasticity of firm markups with respect to the included size measures. $\epsilon_{i,j,t}$ is the residual. Our baseline regressions only use multi-establishment firm-years so that all coefficients are estimated from the same set of firms; otherwise, coefficients on sales and sales per establishment are primarily identified from the vast majority of single-establishment observations.

Table 2: Size-Dependent Markups

	Log markup			
Log sales	0.062^{***}			
	(0.006)			
Log sales per estab.		0.087^{***}		0.087^{***}
		(0.009)		(0.009)
Log number of estabs.			0.021^{**}	0.008
			(0.007)	(0.007)
R^2	0.542	0.545	0.532	0.545
Observations	93,164	93,164	93,164	93,164

Estimated coefficients β from regression (1); robust standard errors clustered at the firm level in parentheses; *: $p < 0.1$, **: $p < 0.01$, ***: $p < 0.001$. Each column includes different size measures, uses multi-establishment firm-years, and includes industry-year fixed effects.

Table 2 reports the results. Column 1 shows a positive, statistically significant relationship between firm sales and markups: 1% higher sales is associated with a 0.062% higher markup.[7] Considered separately, each margin of firm sales also has a positive, statistically significant association with markups (columns 2-3), but a 1% increase in sales via sales per establishment implies four times as big an increase in the markup as a 1% increase via number of establishments. Including both margins together does not affect the coefficient on sales per establishment and reduces the coefficient on number of establishments by two-thirds, rendering it no longer statistically significant at the 10% level (column 4). Thus, firms' average establishment size rather than number of establishments drives the relationship between firm sales and

[7]For comparison, Edmond et al. (2023) estimate an elasticity of 0.031 in US manufacturing data (Table C4 in their online appendix).

markups. Sales per establishment also explains more markup variation than total sales (a higher R^2), and including number of establishments does not improve the fit.

Several alternative specifications in Table 3 confirm our main result: firm markups have a positive, statistically significant relationship with sales per establishment, but no clear relationship with number of establishments. Columns 1-5 use observations with estimated markups above one, consistent with our model. This restriction requires markup levels, which we estimate using industry-year intermediate input cost shares as proxies for output elasticities.[8] This excludes 19% and 13% of single- and multi-establishment observations, respectively, constituting 16% and 9% of single- and multi-establishment firm sales.

Table 3: Size-Dependent Markups, Alternative Specifications

	Baseline	All firms	Firm averages	Firm FEs	Firm FEs + IV	Wage bill markups
Log sales per estab.	0.039***	0.040***	0.036***	0.095***	0.107***	0.249***
	(0.006)	(0.001)	(0.005)	(0.013)	(0.028)	(0.001)
Log number of estabs.	-0.015*	-0.045***	-0.003	0.032**	0.015	-0.046***
	(0.006)	(0.004)	(0.006)	(0.011)	(0.015)	(0.002)
Only markups > 1	✓	✓	✓	✓	✓	
R^2	0.551	0.368	0.566	0.858	0.862	0.634
Observations	80,917	2,790,695	12,340	78,895	73,065	124,343

Estimated coefficients β on both margins of firm sales from different specifications of regression (1); robust standard errors clustered at the firm level in parentheses; *: $p < 0.1$, **: $p < 0.01$, ***: $p < 0.001$. All columns include industry-year fixed effects except 3 (industry effects). Samples are multi-establishment firm-years except columns 2 (all firms) and 3 (firms that are ever multi-establishment). Column 3 uses intermediates-weighted arithmetic means of markups and geometric means of size variables within firms over time. Column 5 instruments size measures with their combined one-year lags. Column 6 use wage bills instead of intermediates in the markup proxy. Further details in the text.

Restricting our baseline regression to firm-years with markups over 1 lowers the coefficient on sales per establishment to 0.039 and turns the coefficient on number of establishments negative (column 1). The coefficient on sales per establishment is

[8] In each 5-digit industry-year, the estimated elasticity of output with respect to intermediates is expenditures on intermediates over total costs, which include wage bills and 0.1 times physical capital stocks, where 0.1 is an approximate user cost as in Hsieh and Klenow (2009). The idea is that firm output is Cobb-Douglas in intermediates, labor, and capital, with constant coefficients across firms within industry-years. See Raval (2023) for a discussion of this approach.

nearly the same if we include single-establishment observations (column 2) or focus on across-firm variation by averaging observations within firms over time (column 3).

We include firm fixed effects to focus on within-firm variation and account for persistent differences across firms in the elasticity of output with respect to intermediates (columns 4-5). The coefficients on both margins rise substantially, yielding the only positive, statistically significant coefficient on number of establishments (column 4). However, this coefficient remains much smaller than that on sales per establishment and falls enough to lose statistical significance when both size measures are instrumented with their combined one-year lags (column 5).[9] This IV specification also addresses the concern that noise in sales affects both the markup proxy and sales per establishment, generating a mechanical correlation.

Finally, replacing intermediates with the wage bill as the variable input in the markup proxy yields a much larger coefficient on sales per establishment (column 6). This may reflect that large establishments have labor market power in addition to product market power, or that adjustment costs slow labor's response to shocks.

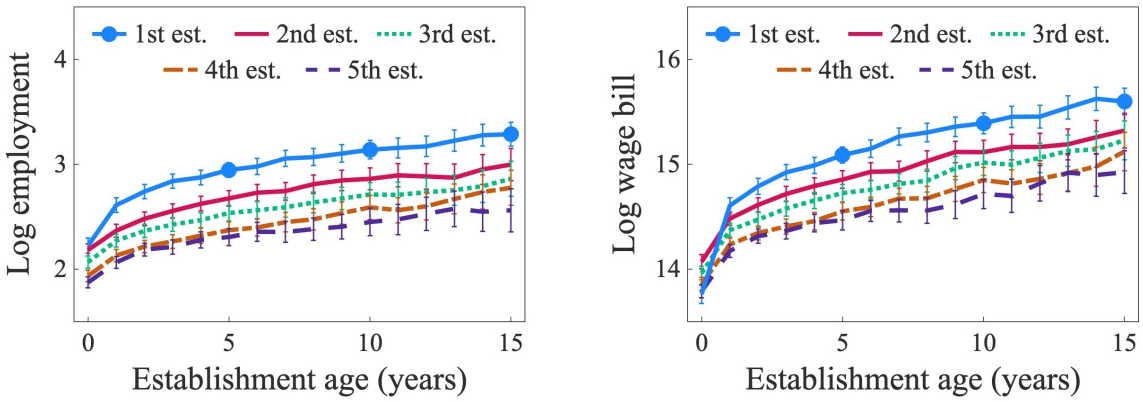
2.3 Successive Establishments Within Firms

We take the preceding results as evidence that market power is increasing in size at the establishment level. To study the implied distortions to firms' extensive margin decisions, we therefore want to know about the size of their *marginal* establishments, i.e., those they will most readily open or close. To this end, we compare establishments within firms based on the order in which they were opened. Since we do not observe establishment sales, we measure size using employment (number of employees) and the wage bill. We find that each successive establishment a firm opens tends to be smaller. We interpret this as firms working down to successively smaller establishment opportunities, so their marginal establishments are relatively small.

[9]The first stage coefficients of sales per establishment on lagged sales per establishment and number of establishments are 0.315 (0.018) and 0.057 (0.012), with robust standard errors clustered at the firm level in parentheses. For number of establishments, they are 0.051 (0.005) and 0.428 (0.009). The second stage coefficient on number of establishments falls to 0.012 and 0.007 if we include two- and three-year lags, whereas the coefficient on sales per establishment remains above 0.089. One explanation is that new establishments are initially small, as shown below in Figure 2, so opening an establishment initially lowers sales per establishment with little effect on the firm markup. This generates a positive coefficient on number of establishments to undo the implied fall in the markup from the positive coefficient on sales per establishment. The IV specification mitigates this issue by focusing on persistent changes in firm sales margins.

Figure 2 plots average log employment and wage bills by age for firms' first five establishments. To account for selection, we only use firms that open at least five establishments. Conditional on survival, later establishments are smaller at all ages.

Figure 2: Firms' Successive Establishments



Mean log establishment employment (left) and wage bills in 2017 SEK (right) by opening order and age, conditional on survival, among firms that open at least five establishments. Vertical bars are 95% confidence intervals.

We quantify this relationship by running the following regression:

$$\ln(\text{establishment size})_{i,j,n,t} = \beta \ln(n) + \text{fixed effects}_{i,j,n,t} + \epsilon_{i,j,n,t}. \tag{2}$$

An observation is an establishment-year pair, indexed by firm i, industry j, establishment order n, and year t, with $n = 1$ for a firm's first establishment, $n = 2$ for its second, and so on. Establishment size is employment or the wage bill. Our baseline specification includes establishment age, firm, and year fixed effects, so the coefficient of interest, β, measures the *within-firm* elasticity of establishment size with respect to opening order net of these effects. Since we include firm fixed effects, we no longer restrict our set of firms to account for selection. $\epsilon_{i,j,n,t}$ is the residual.

Table 4 reports the results. Across all specifications, we estimate a highly statistically significant elasticity of establishment size with respect to opening order of about -0.2. The estimates from our baseline (columns 1 and 4) imply, for example, that a firm's third establishment is about 7.8% smaller than its second, net of establishment age, firm, and year fixed effects. Adding fixed effects for an establishment's municipality-industry (columns 2 and 5) and the age of its firm (columns 3 and 6)

leave the estimates almost unchanged. Therefore, the decline in size across successive establishments within firms is not because firms shift toward municipalities-industries with smaller establishments or because older firms open smaller establishments.

Table 4: Firms' Successive Establishments

	Log employment			*Log wage bill*		
Log establishment order	-0.201***	-0.197***	-0.196***	-0.198***	-0.195***	-0.194***
	(0.021)	(0.020)	(0.020)	(0.020)	(0.019)	(0.020)
Municipality-industry FEs		✓	✓		✓	✓
Firm age FEs			✓			✓
R^2	0.840	0.861	0.861	0.740	0.756	0.756
Observations	6,925,089	6,914,070	6,914,070	6,925,089	6,914,070	6,914,070

Estimated coefficients β from different specifications of regression (2); standard errors clustered at the firm level in parentheses; *: $p < 0.1$, **: $p < 0.01$, ***: $p < 0.001$. All include establishment age, firm, and year fixed effects; check marks indicate additional fixed effects.

We interpret this within-firm ranking as firms working down to successively smaller establishment opportunities. Thus, firms' marginal establishments tend to be smaller than their average. This is distinct from the increasing *across-firm* relationship estimated in Section 2.1 between multi-establishment firms' sales per establishment and number of establishments. Indeed, regressing log sales per establishment on log number of establishments among multi-establishment firm-years, controlling for firm fixed effects in addition to industry-year effects, yields a *negative* estimated elasticity, -0.388, consistent with the new results.[10]

3 Model and Qualitative Results

We now develop and solve our macroeconomic model of firm competition through establishments. We also present our theoretical result, which characterizes the inefficient distortions implied by establishment-level market power. We motivate the model with three broad takeaways from our empirical results, which we match more precisely in our quantitative analysis. First, some firms are at the single-establishment lower bound, whereas others have a relevant extensive margin. Second, firm markups are increasing in the size of firms' establishments, not firms' overall size. Third, each firm

[10]The robust firm-clustered standard error is 0.017, with an R^2 of 0.911, using 91,566 observations.

faces a set of establishment opportunities that vary in size ex-ante. We do not impose any relationship between the size of firms' marginal and average establishments.

Although we use the term establishment throughout, the model and theoretical results also apply to related contexts, such as with multi-product or multi-customer firms and product- or customer-level market power.

3.1 Model

There is a single period. A representative household consumes the numeraire final good, inelastically supplies labor (the only input) in a perfectly competitive market, and owns all firms. A unit mass of firms each controls a measure of establishments, each of which sells a unique establishment-specific variety. Each firm has an initial unit measure of establishments (their first establishment) and chooses which additional establishments to open, if any, as well as how much to sell at each. Each establishment opportunity is characterized by a quality (demand shifter) and opening cost, both known ahead of time. Each firm is characterized by its set of establishment opportunities. Perfectly competitive final good producers aggregate establishment varieties into the final good, which they sell to the household for consumption.

Representative household. The household maximizes final good consumption C subject to its budget constraint, $C \leq W\bar{L}+\Pi$, where W is the wage, $\bar{L}$ is the inelastic labor supply, Π are profits from firms, and the final good price is normalized to 1. Final good consumption C is the measure of welfare.

Final good producers and demand. Perfectly competitive final good producers aggregate a double continuum of varieties from firms' establishments into final good output Y, which they sell as consumption $C = Y$ to the household. Specifically, a unit measure of firms indexed by $i \in [0, 1]$ each operates a measure of establishments indexed by $n \in N(i)$, each of which sells a differentiated variety. Let $y(i, n)$ denote final good producer purchases from firm i's establishment n. Final good output Y is then given implicitly by satisfying the Kimball (1995) aggregator constraint:

$$\int_0^1 \int_{N(i)} \varphi(i,n)\Upsilon(q(i,n))dndi = 1 \qquad\qquad q(i,n) \equiv \frac{y(i,n)}{Y}, \tag{3}$$

where $\varphi(i, n)$ is the quality of firm i's establishment n, $q(i, n)$ is establishment relative

output, and $\Upsilon(\cdot)$ determines the value of purchases from an establishment, scaled by establishment quality. $\Upsilon(\cdot)$ is continuous, strictly increasing, and strictly concave with $\Upsilon(0) = 0$, and has a continuous second derivative on $(0, \infty)$. As such, final good output is increasing in purchases of each variety but with diminishing returns. This aggregation technology is constant returns-to-scale and is CES (constant elasticity of substitution) as a special case, where $\Upsilon(q) = q^{\frac{\bar{\sigma}-1}{\bar{\sigma}}}$ for some $\bar{\sigma} > 1$.

Final good producers take the price $p(i, n)$ of each establishment variety as given and choose demand $y(\cdot, \cdot)$ and final good output Y to maximize profits

$$Y - \int_0^1 \int_{N(i)} p(i, n) y(i, n) dn di$$

subject to the aggregator constraint (3). This yields the inverse demand curve:[11]

$$p(i, n) = \varphi(i, n) \Upsilon'(q(i, n)) D \qquad D \equiv \left(\int_0^1 \int_{N(i)} \varphi(i, n) q(i, n) \Upsilon'(q(i, n)) dn di \right)^{-1}, \quad (4)$$

where $D > 0$ is a demand index that captures the impact of other establishments. An establishment's relative output $q(i, n)$ is decreasing in its price-to-quality ratio, $p(i, n)/\varphi(i, n)$, because $\Upsilon''(\cdot) < 0$. If $\Upsilon'(0) < \infty$, then there is a finite choke price-to-quality ratio $\tilde{p}_{choke}$ above which establishment sales are zero.

Firms, establishments, and production. Each firm $i \in [0, 1]$ has an infinite measure of establishment opportunities $n \in [0, \infty)$. For firm i to produce at establishment n, they must pay a fixed opening cost $\kappa(i, n)$ in labor. The establishment's quality is $\varphi(i, n)$, which shifts its demand as seen above. Establishment output is the labor used for production $l(i, n)$, not including the opening cost, so the marginal cost is the wage W. We add establishment productivity heterogeneity in Section 6.2.

Firm i observes the opening cost and quality at each establishment opportunity and chooses a measurable set $N(i)$ of open establishments as well as the price $p(i, n)$

[11] The first order condition for $y(i, n)$ is $p(i, n) \geq (\lambda/Y)\varphi(i, n)\Upsilon'(y(i, n)/Y)$, holding with equality if $y(i, n) > 0$, where λ is the Lagrange multiplier on the Kimball aggregator constraint. If $y(i, n) = 0$, then without loss of generality, set $p(i, n)$ so it holds with equality. Multiply both sides by $y(i, n)$, integrate across establishments, and plug in that final good producer profits are zero (from perfect competition) to get $Y = \lambda/D$. Plug back into the first order condition to get (4).

and production labor $l(i,n)$ for each open establishment $n \in N(i)$ to maximize profits:

$$\int_{N(i)} (p(i,n) - W) l(i,n) dn - W \int_{N(i)} \kappa(i,n) dn$$

subject to demand (4) and $y(i,n) = l(i,n)$ for each establishment. The first integral is operating profits from open establishments and the second is fixed opening costs.

Single vs. multi-establishment. Each firm can open establishments $n \in [0,1)$ for free, i.e., $\kappa(i,n) = 0$. Without loss of generality, we impose that firms open these establishments so $[0,1) \subseteq N(i)$. We interpret these as a firm's first establishment in our continuous setting. There is no firm entry or exit, so each firm has an initial establishment and then chooses which additional establishments to open. If a firm sets $N(i) = [0,1)$, we call it a "single-establishment" firm; if a firm opens additional establishments, we call it "multi-establishment".

Additional establishment opportunities are not meaningfully ordered by their index $n \geq 1$ so, for example, a firm can open establishments $N(i) = [0,1) \cup [2, 2.6]$. Thus, a firm's measure of open establishments and which establishments constitute that measure are both determined in equilibrium. As such, the correct interpretation is that each firm faces a set of heterogeneous establishment opportunities and opens the best ones. We cannot interpret this as each firm facing a technology where the nth establishment it opens has an exogenous quality and opening cost. In the calibrated model, establishments have a consistent ordering so either interpretation is correct.

Establishment-level market power. A key feature of the inverse demand curve (4) is the elasticity of quantity with respect to price:

$$\frac{-p(i,n)}{y(i,n)} \frac{\partial y(i,n)}{\partial p(i,n)} = \frac{-\Upsilon'(q(i,n))}{q(i,n)\Upsilon''(q(i,n))} \equiv \sigma(q(i,n)). \tag{5}$$

An establishment's demand elasticity is a function $\sigma(q)$ of its relative output q. We assume the aggregator $\Upsilon(\cdot)$ is such that larger establishments face less elastic demand:

Assumption 1. *The demand elasticity $\sigma(q)$ is weakly decreasing for all $q > 0$.*

In this sense, larger establishments have more market power: a lower demand elasticity means an establishment can raise its price with a smaller effect on quantity sold. An establishment's market power depends only on its relative output—determined

by its price-to-quality ratio—and not on its firm's sales at other establishments. This is because there is no special connection in the Kimball aggregator (3) between establishments within the same firm, and because each firm is small in the economy overall and takes aggregates as given. This *establishment-level market power* is motivated by our empirical results on size-dependent markups. By contrast, leaving aside quality, our model would have *firm-level* market power if a firm's contribution to the aggregator constraint were $\Upsilon(\int_{N(i)} q(i,n)dn)$ rather than $\int_{N(i)} \Upsilon(q(i,n))dn$.

Note that even though all establishments enter a single economy-wide aggregator, we need not interpret this to mean that all establishments are in direct competition. Rather, this setup abstracts from variation in the level of competition faced by different establishments, and supposes each establishment faces the same inverse demand curve normalized by its quality. We relax this assumption in Section 6.2.

Firm optimization. We now solve a firm's profit maximization problem. A firm's decisions are independent across establishments because it is small relative to the economy, and establishments are only related through economy-wide aggregates.

On the intensive margin, a firm chooses an establishment's output to equate marginal revenue to marginal cost, which yields the establishment's markup of price over marginal cost:

$$\mu(i,n) = \frac{p(i,n)}{W} = \frac{\sigma(q(i,n))}{\sigma(q(i,n)) - 1}, \tag{6}$$

where $\sigma(q(i,n))$ is the establishment's demand elasticity of quantity with respect to price, defined above in (5). This expression reflects that the firm can sell an additional unit of establishment output at price $p(i,n)$, but to do so must lower the price on inframarginal units, which depresses marginal revenue. More establishment market power, i.e., a lower demand elasticity $\sigma(q(i,n))$, means a wider gap between marginal revenue and the price, which implies a higher profit-maximizing markup.

Assumption 1 of a decreasing demand elasticity means an establishment's equilibrium markup is increasing in its relative output $q(i,n)$. It follows that an establishment's equilibrium relative output and price, as well as the resulting operating profits, are continuous increasing functions of its quality, $q(\varphi)$, $p(\varphi)$, and $\pi(\varphi)$, where

$$\pi(\varphi) = (p(\varphi) - W)q(\varphi)Y.$$

With a finite choke price-to-quality ratio $\tilde{p}_{choke}$, establishments with quality below

$W/\tilde{p}_{choke}$ have zero output because any markup above 1 yields zero demand.

On the extensive margin, firm i opens establishment n if the operating profits $\pi(\varphi(i,n))$ exceed the opening cost $W\kappa(i,n)$. To ensure multi-establishment firms have a well-defined marginal establishment, we make the following technical assumption:

Assumption 2. *In equilibrium, for each firm i, there exists an ordering of their additional establishment opportunities $n \geq 1$—a bijection $\tilde{n}(i,\cdot)$ on $[1,\infty)$—along which the ratio of operating profits to opening costs, $\frac{\pi(\varphi(i,\tilde{n}(i,n)))}{W\kappa(i,\tilde{n}(i,n))}$, is strictly decreasing in n where strictly positive, and is below 1 for sufficiently large n. Moreover, quality $\varphi(i,\tilde{n}(i,n))$ and the opening cost $\kappa(i,\tilde{n}(i,n))$ are continuous in n.*

Assumption 2 implies that each multi-establishment firm has a unique marginal establishment $n^*(i)$ they are indifferent about opening: $\pi(\varphi(i,n^*(i))) = W\kappa(i,n^*(i))$. For each single-establishment firm, define their marginal establishment to be their unique additional establishment opportunity with maximum operating profits relative to opening cost, i.e., $n^*(i) = \tilde{n}(i,1)$. For all firms, let $q_{marg}(i) = q(\varphi(i,n^*(i)))$ be the profit-maximizing relative output at their marginal establishment.

Firm size and markups. Firm-level outcomes are aggregates across their open establishments. Firm i's sales are $S(i) = \int_{N(i)} p(i,n)y(i,n)dn$ and its production labor is $L(i) = \int_{N(i)} l(i,n)dn$. Consistent with our empirical measure in Section 2.2, we define a firm's markup as the production cost-weighted average of its establishment markups, which equals firm sales over production costs: $\mu(i) = S(i)/(WL(i))$.[12]

Aggregation. Final good output is $Y = ZL_p$, where $L_p = \int_0^1 L(i)di$ is aggregate production labor, not including opening costs, and Z is aggregate productivity:[13]

$$Z = \left(\int_0^1 \int_{N(i)} q(i,n)dndi \right)^{-1}. \tag{7}$$

Aggregate productivity depends on the set of open establishments and relative output at each. Quality matters implicitly because higher establishment quality lowers the relative outputs necessary to satisfy the Kimball aggregator (3), which raises Z.

[12] To see this, write $S(i) = \int_{N(i)} p(i,n)y(i,n)dn = \int_{N(i)} \mu(i,n)Wl(i,n)dn$.

[13] To see this, write $Z^{-1} = L_p/Y = \int_0^1 \int_{N(i)} (l(i,n)/Y)dndi$. Use $q(i,n) = l(i,n)/Y$ to get (7).

Equilibrium. The household maximizes consumption C subject to its budget constraint, taking as given wage income $W\bar{L}$ and firm profits Π. Each firm chooses their set of open establishments $N(i)$ and the price $p(i,n)$ and labor $l(i,n)$ at each establishment to maximize profits, taking as given aggregate output Y, the wage W, and the demand index D. Final good producers choose establishment demand $y(\cdot,\cdot)$ and aggregate output Y to maximize profits, which must be zero due to perfect competition, taking as given establishment prices $p(\cdot,\cdot)$.

The following markets clear. Household consumption equals aggregate output: $C = Y$. Demand for each establishment variety equals output, which is the labor used to produce it: $y(i,n) = l(i,n)$. Total labor used for production and fixed opening costs equals the inelastic supply: $\int_0^1 L(i)di + \int_0^1 \int_{N(i)} \kappa(i,n)dndi = \bar{L}$.

Find an equilibrium as follows. Given aggregate output Y and the demand index relative to the wage D/W, use inverse demand (4) and the optimal markup (6) to get an establishment's relative output and its operating profits relative to the wage as functions of its quality, $q(\varphi)$ and $\pi(\varphi)/W$. The latter implies each firm's set of open establishments $N(i)$. Then, check whether the Kimball aggregator constraint (3) holds and the labor market clears. Hence, there are two aggregate variables (Y and D/W) to satisfy two equations (Kimball aggregator constraint and labor market clearing). The remaining equilibrium variables follow.

3.2 Planner Problem and Distortions in the Market Equilibrium

We now characterize how establishment-level market power distorts firms' choices of which establishments to open and how much to produce at each. Section 5 uses the calibrated model to solve for the efficient allocation, compute the costs of these distortions, and study their implications for firm size-dependent policy.

A planner chooses each firm's set of open establishments and production at each establishment to maximize aggregate output Y, subject to the Kimball aggregator constraint (3), firms' establishment opening and production technologies, and the fixed labor supply $\bar{L}$. The solution to the planner problem is the efficient allocation.

The market allocation can be inefficient due to misallocation of the fixed labor supply. We can raise aggregate output by shifting labor from uses with a lower marginal social value to uses with a higher marginal social value. To characterize these opportunities, let $V_{prod}(i,n)$ be the marginal social value of labor used to produce at firm i's open establishment n, and let $V_{estab}(i)$ be the marginal social value of labor used

to open and produce at firm i's marginal establishment. Formally, hold fixed the allocation of labor from the market equilibrium. If we use an additional infinitesimal quantity Δ of labor to increase production at firm i's establishment n, then aggregate output increases by $V_{prod}(i,n)\Delta$. If we use Δ labor to open marginal establishments at firm i and produce their market equilibrium output $q_{marg}(i)Y$, then aggregate output increases by $V_{estab}(i)\Delta$. This means opening $\Delta/(\kappa(i,n^*(i)) + q_{marg}(i)Y)$ establishments because each has fixed opening cost $\kappa(i,n^*(i))$ and uses $q_{marg}(i)Y$ labor for production. We can now state our result.

Theorem 1. *If the establishment demand elasticity $\sigma(q)$ is constant in relative output q, then the market equilibrium is efficient.*

If $\sigma(q)$ is strictly decreasing for all $q > 0$, then the following are true:

1) **Intensive margin.** *At all producing establishments, $V_{prod}(i,n) = W\mu(i,n)$, where the markup $\mu(i,n)$ is strictly increasing in relative output $q(i,n)$.*

2) **Extensive margin.** *At all firms, $V_{estab}(i) \leq W\frac{\Upsilon(q_{marg}(i))}{q_{marg}(i)\Upsilon'(q_{marg}(i))}$, with equality if firm i is multi-establishment. The right-hand side is strictly increasing in $q_{marg}(i)$, and therefore so is $V_{estab}(i)$ among multi-establishment firms.*

3) **Intensive vs. extensive.** *For any firm i and producing establishment n at firm j, $V_{estab}(i) < V_{prod}(j,n)$ if $q_{marg}(i) \leq q(j,n)$. By continuity, we also have $V_{estab}(i) < V_{prod}(j,n)$ for $q(j,n)$ less than but sufficiently close to $q_{marg}(i)$.*

These imply an inefficient market equilibrium if there is a nondegenerate distribution of establishment relative output or if there are multi-establishment firms.

Proof. See Appendix A.1. ■

Under CES demand, we cannot improve on the market allocation. On the other hand, if the demand elasticity is strictly decreasing in establishment size, then the theorem identifies three ways to raise welfare. First, the familiar size-dependent markup distortion to the intensive margin: the marginal social value of establishment production labor is proportional to the markup, so we can increase welfare by shifting production from smaller, lower markup establishments to larger, higher markup ones. Second, the extensive margin is similarly distorted by size: welfare rises by closing

smaller marginal establishments at one firm and using the labor to open larger ones at another multi-establishment firm. Third, comparing the two margins, the labor saved by closing a marginal establishment is more valuable if used instead to expand production at a weakly larger establishment, or even a somewhat smaller one.

A firm's markup, its production cost-weighted average establishment markup, therefore captures the marginal social value of expanding its existing establishments but not of expanding by opening new establishments. This is both because the intensive and extensive margins are distorted differently even at the same establishment size, and because a firm's marginal establishment size may differ from its average establishment size. Indeed, consistent with our empirical results, firms' marginal establishments are relatively small in our calibrated model, making it especially valuable to close them in order to expand production at remaining establishments.

To understand these results, compare the marginal social value of each use of labor to the marginal private value. The market equilibrium equalizes the additional revenue per unit of labor to the wage across all active uses, so labor has a higher marginal social value where the firm captures less of that value through revenue.

On the intensive margin, the marginal social value of production labor at an establishment with quality φ and relative output q is

$$\varphi \Upsilon'(q) D, \tag{8}$$

which is the establishment's price from its inverse demand (4). Marginal revenue is this price multiplied by $1 - 1/\sigma(q)$, which reflects that due to market power, i.e., a finite demand elasticity $\sigma(q)$, selling an additional unit requires cutting the price on inframarginal units. As the demand elasticity falls with establishment size, marginal revenue falls relative to the price, so the private value of increasing establishment production labor falls relative to the social value. The market equilibrium equates marginal revenue to the wage, so it follows that the markup and the marginal social value of production labor are higher at larger establishments (item 1).

On the extensive margin, the marginal social value of labor used to open and produce at an establishment with quality φ, opening cost κ, and output qY is

$$\frac{\varphi \Upsilon(q) DY}{\kappa + qY}. \tag{9}$$

The numerator is the increase in aggregate output from adding the establishment's total output, given by the area under its inverse demand curve, and the denominator is the labor required for the opening cost and production. For the private value per unit of labor, the numerator is replaced by establishment revenue, $q\varphi\Upsilon'(q)DY$. Revenue is quantity times price, the smaller rectangle under the inverse demand curve that values all units of establishment output at the social value of the marginal unit.

Now, the private-to-social value ratio for an establishment's total output is

$$\frac{q\Upsilon'(q)}{\Upsilon(q)} = \frac{\int_0^q \Upsilon'(\tilde{q}) \left(1 - \frac{1}{\sigma(\tilde{q})}\right) d\tilde{q}}{\int_0^q \Upsilon'(\tilde{q}) d\tilde{q}},$$

an average of the private-to-social value ratio across each unit of its output. The demand elasticity falls with establishment size, so that ratio is lower with each successive unit. Thus, the average ratio, $q\Upsilon'(q)/\Upsilon(q)$, falls with q and is above the ratio for the last unit, $1 - 1/\sigma(q)$. The market equates a multi-establishment firm's total revenue per unit of labor at its marginal establishment to the wage, just as it equates the marginal revenue from expanding an establishment's production labor to the wage. Therefore, the social value per unit of labor from opening a marginal establishment is increasing in establishment size (item 2), and is less than the marginal social value of expanding production labor at an establishment of the same size (item 3).

Finally, returning to CES demand, market power is constant in establishment size, so revenue captures the same share of social value across all activities. As such, there is no misallocation and the market equilibrium is efficient.

4 Quantifying the Model

We now calibrate the model to match our data on services industries in the Swedish economy from 1997 to 2017. We then illustrate features of the equilibrium and confirm key model predictions in the data. In the following section, we use the calibrated model to study misallocation and firm size-dependent policy.

4.1 Calibration

We first impose functional forms for the distribution of establishment quality and opening costs within and across firms, and for the Kimball aggregator function. To

keep the analysis of firm size-dependent policy tractable, we restrict exogenous firm heterogeneity to a single variable, "baseline quality" or "firm quality", which shifts the quality of all a firm's establishment opportunities. Higher quality firms are larger, and those above a baseline quality cutoff operate multiple establishments.

Establishment quality and opening costs. A firm's establishments are ordered so that in any equilibrium, if firm i opens a measure $|N(i)|$ establishments, it opens establishments $n \in [0, |N(i)|]$. Specifically, firm i's baseline quality $\bar{\varphi}(i)$ is drawn from a Pareto distribution with minimum 1 and tail parameter $\omega > 0$. Then its establishment qualities and opening costs are

$$\varphi(i,n) = \begin{cases} \bar{\varphi}(i), & 0 \leq n < 1, \\ \bar{\varphi}(i) n^{-\rho}, & n \geq 1, \end{cases} \qquad \kappa(i,n) = \begin{cases} 0, & 0 \leq n < 1, \\ \bar{\kappa} n^{\theta}, & n \geq 1. \end{cases}$$

Thus, each firm has an initial unit measure of identical establishments that are free to open with quality equal to baseline. Additional establishments have progressively lower quality, with elasticity $\rho > 0$, consistent with our evidence that successive establishments within firms are smaller.

The parameter $\bar{\kappa} > 0$ is the fixed opening cost at a firm's first marginal additional establishment, and θ governs how this cost changes with each successive establishment. We allow increasing ($\theta > 0$), constant ($\theta = 0$), or decreasing ($\theta < 0$) opening costs, subject to $\theta > -\rho$, which ensures that the ratio of establishment operating profits to opening costs, $\pi(\varphi(i,n))/(W\kappa(i,n))$, is strictly decreasing in n wherever positive and is eventually below 1, as in Assumption 2. This yields an interior extensive margin optimum and the desired ordering of establishment opportunities by n.

Demand. We use the Klenow and Willis (2016) specification for the derivative of the Kimball aggregator function:

$$\Upsilon'(q) = \frac{\bar{\sigma} - 1}{\bar{\sigma}} \exp\left(\frac{1 - q^{\epsilon/\bar{\sigma}}}{\epsilon}\right),$$

where $\bar{\sigma} > 1$ and $\epsilon \geq 0$. This implies the establishment demand elasticity

$$\sigma(q) = \bar{\sigma} q^{-\epsilon/\bar{\sigma}}.$$

The parameter $\bar{\sigma}$ is the demand elasticity at establishment relative output $q = 1$, and $\epsilon/\bar{\sigma}$ determines how rapidly demand becomes less elastic with establishment size. If $\epsilon > 0$, then larger establishments face less elastic demand and set higher markups. The limiting case $\epsilon = 0$ yields CES demand and constant markups. We complete the specification by setting $\Upsilon(0) = 0$ so non-producing establishments have no effect.

Calibration strategy. We normalize the inelastic labor supply $\bar{L}$ so that aggregate output is $Y = 1$. We jointly calibrate the remaining six parameters to exactly match six moments. Guided by Theorem 1, we choose targets that discipline the importance of firms with an active extensive margin, the sizes of open and marginal establishments across firms, and the relationship between establishment size and markups. Table 5 lists the moments, parameters, and their calibrated values, both for this main calibration and for the alternative specifications in Section 5.3.

Table 5: Calibration Targets and Calibrated Parameter Values

Moment	**Data**	**Model**
multi-estab. firm sales share	0.49	0.49
multi-estab. firm frequency	0.027	0.027
multi-estab. firm relative establishment sales	2.92	2.92
wage bill decline across successive estabs.	0.20	0.20
cost-weighted average markup	1.63	1.63
multi-estab. log(markup) on log(sales per estab.)	0.087	0.087

Parameter	**Baseline**	*Uniform pricing*	*Low markup*	*Lognormal*
ω (baseline quality parameter)	8.19	9.03	11.77	0.32
$\bar{\kappa}$ (estab. opening cost shifter)	0.45	0.43	0.29	0.84
θ (estab. opening cost elasticity)	0.086	0.106	0.084	0.020
ρ (estab. quality elasticity)	0.107	0.086	0.078	0.103
$\bar{\sigma}$ (demand elasticity at $q = 1$)	2.52	2.53	4.51	2.55
$\epsilon/\bar{\sigma}$ (demand super elasticity)	0.14	0.15	0.29	0.13

ω is the baseline quality tail parameter in all cases except "lognormal", where it is the standard deviation of log baseline quality. Calibration targets are the same in all cases except "low markup", where the cost-weighted average markup is 1.29.

First, we match the sales share of multi-establishment firms and the share of firms that are multi-establishment. The sales share captures the aggregate importance of firms with an active extensive margin. Together with multi-establishment firms'

frequency, it also pins down their average relative sales. Since they are the largest firms in the model, this disciplines the tail of the firm-size distribution. As such, these moments are particularly informative about the baseline quality tail ω and the opening cost shifter $\bar{\kappa}$, which governs the threshold for being multi-establishment.

Second, we match average sales per establishment across multi-establishment firms' establishments relative to all establishments. This disciplines intensive margin distortions. Moreover, given the first two targets, we therefore exactly match the decomposition of multi-establishment firms' large sales into large establishments versus many establishments. This is particularly informative about the curvature of opening costs: a higher θ raises the cost of opening many establishments, so multi-establishment firms' sales shift from many establishments to larger establishments.

Third, we match the within-firm elasticity of establishment wage bills with respect to establishment order by running the same regression in our model (with only firm fixed effects) as for our empirical results in Table 4. Since we match the average establishment size and number of establishments among multi-establishment firms, this pins down the average size of their *marginal* establishments, which are central to extensive margin distortions. This target mostly informs ρ, the rate at which quality declines across successive establishments within firms.

Finally, we match the level of markups and the markup-size relationship. These discipline the average demand elasticity, $\bar{\sigma}$, and its elasticity with respect to establishment size, $\epsilon/\bar{\sigma}$, which map the preceding moments into distortions. For the level of markups in the model, we use their production labor-weighted average. We match the midpoint between the intermediates-weighted average of 1.97 estimated in our data and a sales-weighted average of 1.29 estimated by De Loecker and Eeckhout (2025) across all Swedish industries.[14] We get similar results in Section 5.3 using 1.29. For the markup-size relationship in the model, we regress multi-establishment firms' log markups on their log sales per establishment. We match our baseline estimated elasticity in Table 2, which sits between our alternative specification estimates.

An important assumption is that all establishments are valued by the same function $\Upsilon(\cdot)$. We use markups of firms with different sized establishments to discipline

[14]We describe our markup estimation in Section 2.2. The model sales-weighted average markup is 1.64, nearly identical to the cost-weighted average. We include the low estimate from De Loecker and Eeckhout (2025) in part because they find that the cost-share approach we use overestimates markups. Christopoulou and Vermeulen (2012) estimate a sales-weighted average markup of 1.56 in services industries across eight euro-area countries, not including Sweden, over 1981-2004.

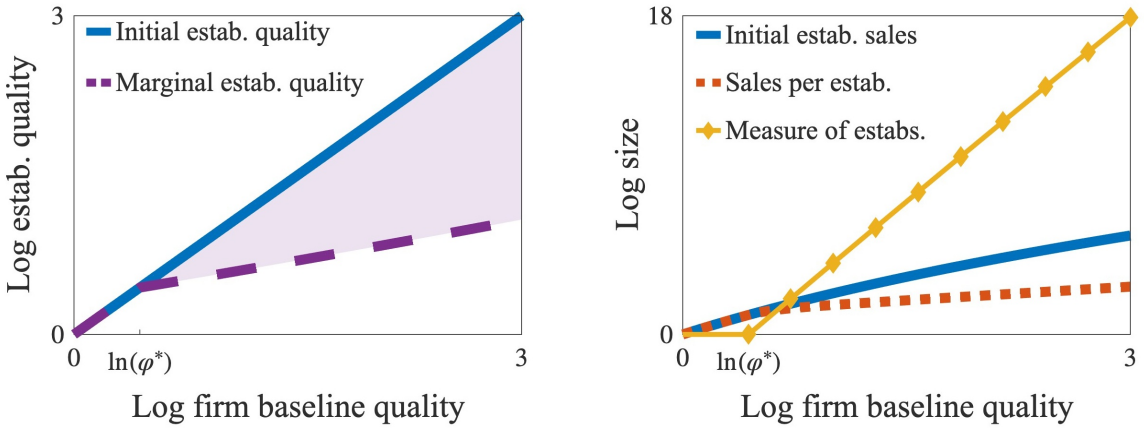
the shape of $\Upsilon'(\cdot)$. Along with $\Upsilon(0) = 0$, we can then compute the total social value $\Upsilon(q)$ of an establishment with high relative output q even though that establishment's markup is only directly informative about the local value $q\Upsilon''(q)/\Upsilon'(q)$.

4.2 Equilibrium Illustration and Testing Model Predictions

We now illustrate key features of the calibrated market equilibrium and compare several model predictions to our Swedish services data.

The left panel of Figure 3 plots each firm's range of open establishment qualities, where firms are indexed by their baseline quality. Firms with baseline quality above φ^* are multi-establishment, and higher quality firms open more establishments. A multi-establishment firm opens successively lower quality establishments until its marginal establishment, its smallest, has operating profits equal to opening cost. Across firms, marginal establishment quality rises with baseline quality because opening costs are increasing in number of establishments ($\theta > 0$), so firms with more establishments require higher marginal establishment operating profits to compensate.

Figure 3: Firms' Establishments and Sales



Left: the range of open establishment qualities between a firm's initial, $n \in [0, 1)$, and marginal, $n^*(i)$, establishments, as a function of firm baseline quality. Right: initial and average establishment sales, and measure of establishments, as functions of baseline quality, normalized by the lowest quality firm. Firms are multi-establishment above the cutoff φ^*.

The right panel of Figure 3 shows the implications for the two margins of firm sales. Higher quality firms have higher quality, larger initial establishments. Sales per establishment does not rise as steeply with firm quality because higher quality

firms open more establishments, bringing down average establishment quality. As a consequence, differences in sales among larger firms increasingly reflect their measure of establishments rather than their sales per establishment, as in our data (Figure 1).

Table 6 tests untargeted features of firm sales and its two margins in the model against the data. First, the model closely matches the relationship between sales per establishment and number of establishments among multi-establishment firms: regressing the former on the latter in logs yields an elasticity of 0.150, compared with 0.143 in the data controlling for industry-year fixed effects (Section 2.1). The calibration only targets the average values of these two sales margins, not their relationship.

Next, we compare the largest firms' sales shares in the model with empirical shares across all services firms, and with sales-weighted averages of within-industry shares. The model understates the share of the largest 5% and overstates that of the largest 0.1%, implying a thicker upper tail. Using a lognormal baseline quality distribution yields a thinner tail without substantively changing our results (Section 5.3).

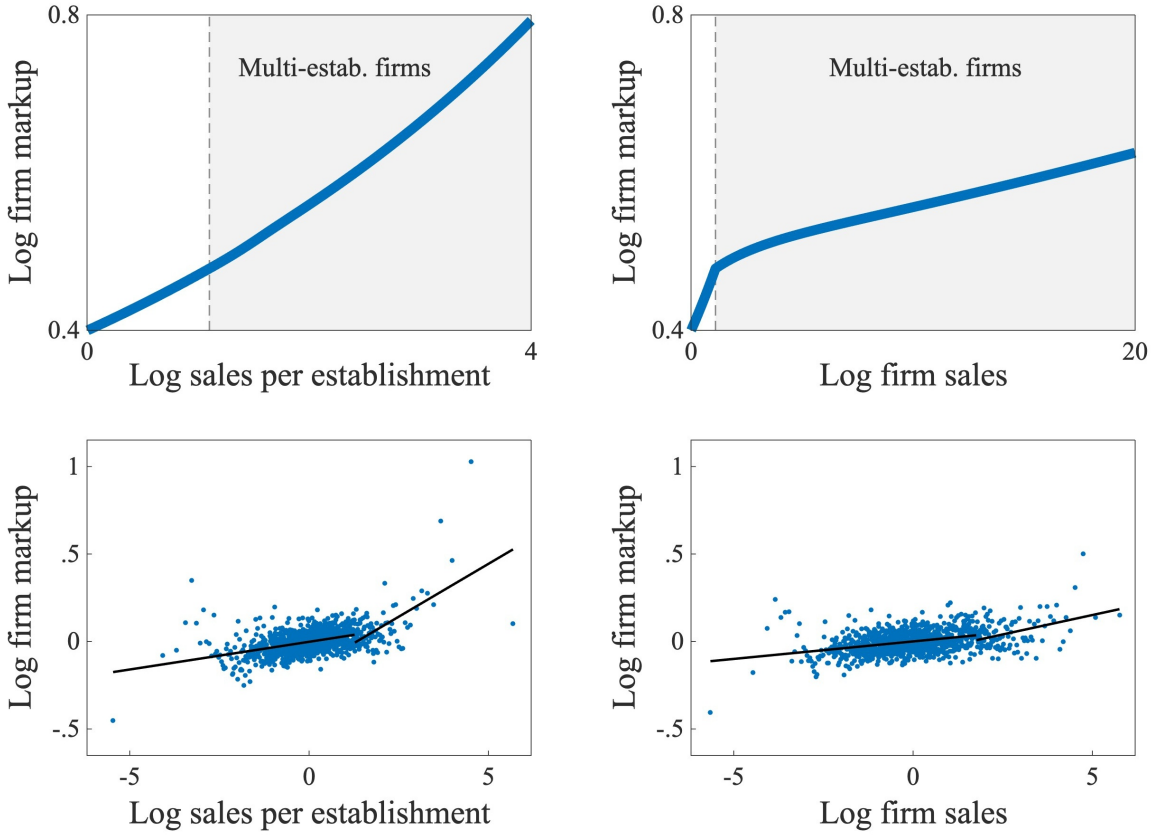
Table 6: Untargeted Sales Moments in the Model vs. Data

	Elasticity between margins	**Sales concentration**		
		Top 5%	Top 1%	Top 0.1%
Model	0.150	51.1%	45.4%	37.7%
Data (economy-wide)		76.8%	58%	33%
Data (within-industries)	0.143	67.0%	45.4%	27.3%

Finally, Figure 4 shows a common pattern in how the two margins of firm sales shape firm markups, cost-weighted average establishment markups, in the model and data. In the model (top panels), log firm markups are convex in log sales per establishment and also increasing in log firm sales, but with a much flatter and almost linear relationship among multi-establishment firms.[15] This reflects that larger firms mostly have more establishments rather than larger establishments, which raises their sales but not their market power. The bottom panels confirm this untargeted prediction in the data. Among multi-establishment firms, the within-industry-year relationship between log firm markups and log sales per establishment steepens in the upper tail, whereas the relationship with log firm sales is flatter and nearly linear.

[15]The model sales per establishment panel has a higher log baseline quality upper bound (5.6) than the sales panel (3), but exhibits less size variation. Sales per establishment varies much more in the data where it does not map one-for-one into sales. We thus keep the comparison qualitative.

Figure 4: Size-Dependent Markups in the Model (top) vs. Data (bottom)



Top: model firm markups as a function of sales per establishment (left) and firm sales (right), with size measures normalized by the lowest quality firm. Bottom: multi-establishment firm-years from the Swedish services data grouped into 1000 equally sized bins, ranked by sales per establishment (left) and firm sales (right), normalized by industry-year means. Each dot plots the average normalized log markup within a bin against the average normalized log size. Solid lines are best linear fits for the lower 90% and upper 10% of bins.

5 Quantitative Results

We now solve the planner problem and study optimal firm size-dependent policy in the calibrated model. We highlight the importance of a conflict between what is socially beneficial along the intensive vs. extensive margins of firm production. Larger firms set higher markups, so they underproduce along the intensive margin, which calls for a subsidy. But the largest firms have multiple establishments. This mostly implies overproduction and calls for a tax because firms overvalue expansion along

the extensive margin, especially given their relatively small marginal establishments.

5.1 The Planner Problem

To solve the planner problem, equalize the marginal social value of production labor across all producing establishments and open an establishment if its social value per unit of labor exceeds this constant value. These social values are given by (8) and (9) from our discussion of Theorem 1, evaluated at the efficient allocation.

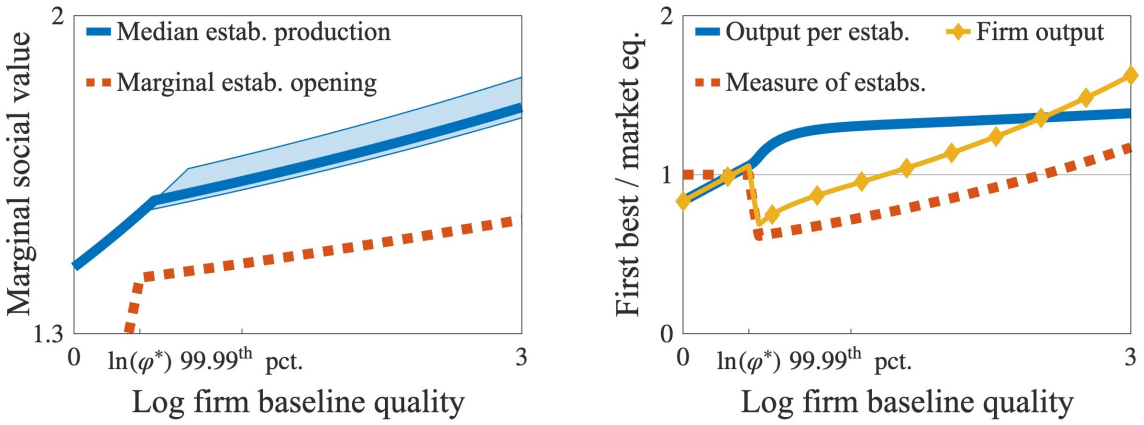
First best allocation. The planner reallocates the fixed labor supply to maximize aggregate output. The left panel of Figure 5 plots the drivers of this reallocation: the marginal social values of uses of labor in the market equilibrium, $V_{prod}(i, n)$ and $V_{estab}(i)$, characterized in Theorem 1. On the intensive margin, the value of using labor to expand existing establishments (solid line) is higher at larger firms because they set higher markups. It varies little across establishments within firms, shown by the shaded area for each firm size. On the extensive margin, the value of using labor to open firms' marginal establishments (dotted line) also rises with firm size as larger firms have larger marginal establishments more undervalued by revenue.

A significant opportunity to improve welfare exists between the two margins (solid line vs. dotted line) because opening a firm's marginal establishment has a lower social value per unit of labor than expanding production even at substantially smaller firms' establishments. It is therefore valuable to close establishments at multi-establishment firms in order to increase production at the establishments of almost any firm.

The right panel of Figure 5 shows the resulting efficient reallocation of labor. Output per establishment rises at large single-establishment firms and at all multi-establishment firms. Establishments close at all multi-establishment firms except the very largest one in ten million. The measure of additional establishments falls by 26%, but fixed opening costs fall by only 8.7% because closures are concentrated at smaller multi-establishment firms with lower marginal opening costs.

The net effect on total firm output is non-monotonic in firm size and markups. The largest single-establishment firms expand, but 93% of the even larger, higher markup multi-establishment firms contract because establishment closures outweigh increases in output per establishment. Indeed, the largest percentage declines in output occur among firms between the 98th and 99.85^{th} percentiles of the firm sales distribution. Thus, a high markup does not imply that a firm should expand overall.

Figure 5: Solving the Planner Problem



Left: $V_{prod}(i,n)$ at firm i's median quality establishment (90th to 10th percentiles in the shaded area) and $V_{estab}(i)$ as functions of baseline quality. Right: output per establishment, measure of establishments, and total firm output in the first best relative to the market equilibrium, as functions of baseline quality. The 99.99^{th} percentile largest firm is indicated.

Costs of misallocation. The first best raises aggregate output by 1.52% relative to the market equilibrium. Recalling that aggregate output is $Y = ZL_p$, these gains come from a 0.43% fall in aggregate productivity Z and a 1.96% increase in production labor L_p. This understates the costs of misallocation in the sense that it does not account for amplification through intermediate inputs and capital accumulation.[16]

We decompose misallocation by sequentially relaxing constraints on the planner. Holding each firm's set of establishments and production labor fixed from the market equilibrium, reallocating production within firms raises output by only 0.03%, reflecting the minimal within-firm variation in the marginal social value of establishment production labor we saw in Figure 5. Allowing production labor also to move across firms increases the total gain to 0.85%. Permitting the set of establishments to change as well delivers the full 1.52% first best improvement over the market equilibrium.

[16]For example, in a similar model with firm-level market power, Edmond et al. (2023) implement a size-dependent policy that eliminates misallocation and leaves the aggregate labor supply unchanged. Initially, aggregate productivity rises by 1.75%. Accounting for the transition to a new steady state, welfare rises by the equivalent of a 5.58% permanent increase in consumption. Their version with quality rather than productivity heterogeneity is in online appendix E.1.

5.2 Firm Size-Dependent Policy

We now study the shape of optimal firm size-dependent policy and ask how well it can alleviate misallocation. Our focus on policy that conditions only on total firm sales is relevant for understanding the effects of various real-world firm size-dependent taxes and regulations, as well as changes in the economic environment more generally.[17] Moreover, it may be difficult to reliably observe establishment-level sales in order to implement a richer policy. This is especially true in related contexts such as with multi-product or multi-customer firms and product- or customer-level market power.

The key tension is that firm size-dependent policy cannot separately incentivize the intensive vs. extensive margins, i.e., production at existing establishments vs. expansion through new establishments. This limits the effectiveness of policy and makes the optimal schedule non-monotonic: large single-establishment firms face lower marginal tax rates than smaller firms and than nearly all larger, higher markup multi-establishment firms. Adding a flat tax on opening establishments resolves this tension, restoring monotonicity and eliminating nearly all misallocation.

Formally, the policy is a marginal relative sales tax schedule $\tau(\cdot)$. Firm i's nominal sales relative to aggregate output is $S(i)/Y$ and its tax bill is $Y \int_0^{S(i)/Y} \tau(x)dx$, which is normalized to 0 for a firm with zero sales. Thus, firm i's marginal revenue from any margin of production is multiplied by $1 - \tau(S(i)/Y)$, where a negative value of $\tau(\cdot)$ represents a marginal subsidy. The policy is announced before firms choose their allocations and is taken as given in the market equilibrium. Net tax revenues, positive or negative, are transferred to the household lump sum.

For all policy exercises, we also normalize $\tau(\cdot)$ to keep the labor share $W\bar{L}/Y$ fixed at its equilibrium value without policy. Specifically, multiplying $1-\tau(\cdot)$ by a positive constant scales the equilibrium wage proportionately without changing the real allocation. We therefore must choose this constant, which determines where marginal rates are positive or negative, but not their ranking across firms or any non-monotonicity.[18]

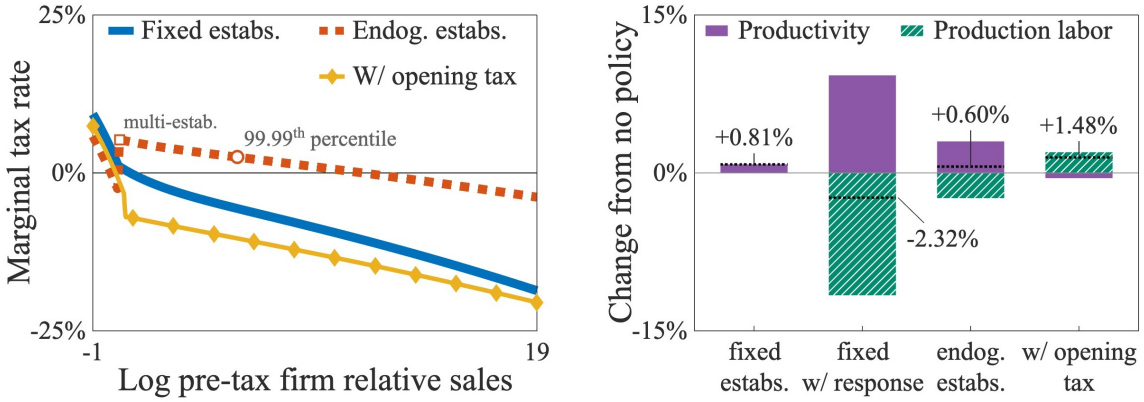
[17] OECD (2015) surveys firm size-dependent tax policies in G20 countries. Garicano et al. (2016) document firm employment-dependent regulations in France. Almunia and Lopez-Rodriguez (2018) study a policy in Spain that imposed stricter tax enforcement on firms with revenues over €6 million. As a recent example of firm size-dependent economic shifts, artificial intelligence may favor large firms if it increases fixed cyber security costs, or hurt them if it makes them targets.

[18] This normalization would also keep labor supply fixed under the commonly used household preferences for consumption and labor, $\ln(C) - \psi L^{\eta}$, with $\psi > 0$ and $\eta > 1$.

Fixed extensive margin. To illustrate the role of the intensive margin, we first design policy to maximize output holding each firm's set of open establishments fixed from the market equilibrium without policy. Optimal policy equates $\bar{V}_{prod}(i)$ across all firms, where $\bar{V}_{prod}(i)$ is the average marginal social value of production labor, $V_{prod}(i,n)$, across firm i's establishments, weighting each establishment by its production labor response to a change in firm i's marginal tax rate $\tau(S(i)/Y)$.

The left panel of Figure 6 displays the optimal marginal tax schedule $\tau_{fixed}(\cdot)$, labeled "fixed estabs". Larger firms face lower tax rates because they set higher markups. The right panel shows the effects of implementing this policy. Aggregate productivity and output rise by 0.81%, eliminating almost all of the 0.85% losses from misallocation given the fixed set of establishments. Thus, little is lost from conditioning only on firm sales despite markup heterogeneity within firms.

Figure 6: Optimal Firm Size-Dependent Policies



Left: optimal firm size-dependent marginal tax schedules $\tau_{fixed}(\cdot)$ "fixed estabs.", $\tau_{endog}(\cdot)$ "endogenous estabs.", and $\tau_{flat}(\cdot)$ "w/ opening tax"; the equilibrium multi-establishment firm cutoff and 99.99th percentile largest firm are indicated for $\tau_{endog}(\cdot)$. Right: log changes in aggregate productivity, production labor, and output from implementing each policy, including the effects of $\tau_{fixed}(\cdot)$ in the equilibrium with extensive margin responses.

Endogenous extensive margin. We next implement the fixed extensive margin policy, $\tau_{fixed}(\cdot)$, allowing the set of open establishments to respond in equilibrium.[19] The right panel of Figure 6 shows that output *falls* by 2.32% relative to the equilib-

[19]We impose a marginal tax rate lower bound of -100% when solving for each optimal policy, which has negligible effects except that it limits a huge fall in output in this exercise.

rium without policy because productivity rises by 9.71% but at the cost of a 10.97% drop in production labor. That is, policy designed to undo intensive margin markup distortions oversubsidizes large firms, leading them to open too many establishments.

We now design policy to maximize output taking into account firms' full responses. Optimal policy equates $(1-\nu(i))\bar{V}_{prod}(i)+\nu(i)V_{estab}(i)$ across all firms. A fall in firm i's marginal tax rate $\tau(S(i)/Y)$ raises production at its existing establishments, with marginal social value $\bar{V}_{prod}(i)$ as before, and may also lead to the opening of new establishments, with marginal social value $V_{estab}(i)$. The weight $\nu(i)$ is the share of additional labor used for opening and producing at new establishments.[20]

Figure 6 plots the optimal marginal tax schedule $\tau_{endog}(\cdot)$, labeled "endog. estabs.", and the resulting aggregate effects. The policy is non-monotonic in firm size and markups. Marginal tax rates are falling in size for most single-establishment firms, as before, because larger firms set higher markups. Near the transition to multi-establishment status, however, they steadily rise to prevent firms from opening additional establishments. Tax rates then remain higher for nearly all multi-establishment firms to limit extensive margin expansion. Ultimately, the largest one in a billion firms still face the lowest tax rates because they set such high markups and their large marginal establishments have sufficiently high social value.

Output rises by 0.60%, compared to 0.81% from optimal policy with a fixed set of establishments. Allowing for extensive margin adjustment thus yields higher potential gains (1.52% vs. 0.85%), yet lowers the gains from optimal firm size-dependent policy. This reflects a tension between intensive and extensive margin distortions.

Tax on opening establishments. Finally, to address this tension, we add a flat tax $\bar{\tau}$ on fixed opening costs, which we jointly choose along with the firm size-dependent policy $\tau_{flat}(\cdot)$ to maximize output taking into account firms' responses along both margins. Firm i's total tax bill is $Y\int_0^{S(i)/Y}\tau_{flat}(x)dx+\bar{\tau}W\int_{N(i)}\kappa(i,n)dn$.

The optimal tax on opening costs is 48.0%. Figure 6 shows the size-dependent marginal tax schedule $\tau_{flat}(\cdot)$, labeled "w/ opening tax", and the aggregate effects. Monotonicity is restored with larger, higher markup firms again facing lower tax rates. Output rises by 1.48%, achieving almost all the 1.52% first best gains. Thus, a

[20]For some firms indifferent about becoming multi-establishment, labor attracted by a tax cut has too little social value (with $\nu(i)>0$), but labor released by a tax increase has too much social value (with $\nu(i)=0$). Optimal policy therefore keeps these firms at the multi-establishment boundary.

flat extensive margin tax is mostly able to resolve the tension between intensive and extensive margin distortions, even though this tension is not constant across firms.

5.3 Alternative Specifications

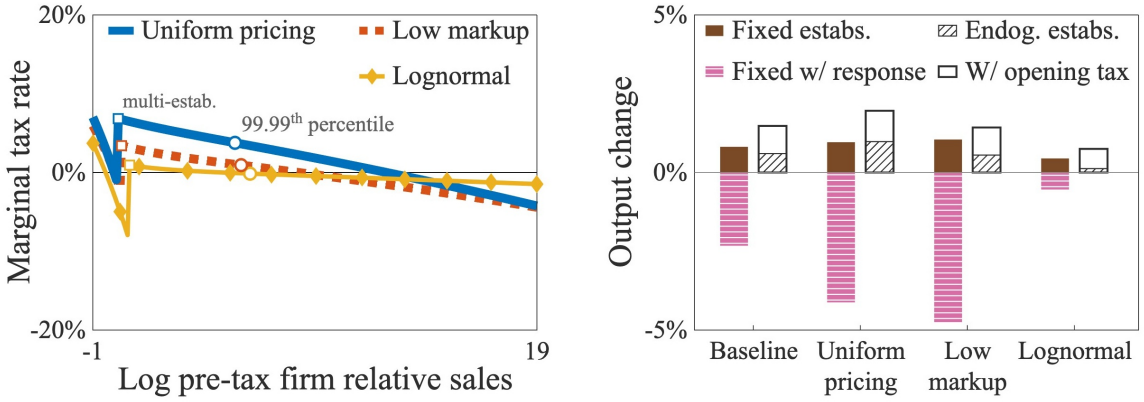
Our results are robust to three alternative specifications. First, we impose uniform pricing, so each firm must set a single price across all their establishments. We provide more detail and extend our theoretical results to this setting in Section 6.1. Second, returning to establishment-specific pricing, we recalibrate the model to match a lower cost-weighted average markup of 1.29 based on De Loecker and Eeckhout (2025). Third, we draw baseline quality from a lognormal rather than Pareto distribution, and recalibrate the model to match our original targets. In this case, the firm sales distribution has a thinner rather than thicker tail compared to the data.[21] The calibrated parameters in each case are listed in Table 5 above in Section 4.1.

For the sake of brevity, we only show the headline results for each alternative specification. The left panel of Figure 7 illustrates the optimal firm size-dependent marginal tax schedule $\tau_{endog}(\cdot)$ that maximizes output when firms respond along both margins. Marginal tax rates are non-monotonic in firm sales and markups, as before.

The right panel plots the aggregate output effects of our various optimal firm size-dependent policies. The first pair of bars for each specification show the gains from implementing $\tau_{fixed}(\cdot)$, optimal policy given a fixed extensive margin, and the losses if the set of open establishments then responds. The second pair of bars show the gains from implementing $\tau_{endog}(\cdot)$, optimal policy taking into account firms' responses along both margins, and $\tau_{flat}(\cdot)$, which is chosen jointly with a flat tax on fixed opening costs. As before, allowing the set of open establishments to adjust always raises potential gains, but not the gains from optimal firm size-dependent policy without the flat tax. Including the flat tax resolves the tension between the intensive and extensive margins, and eliminates at least 96% of total losses from misallocation.

[21] The sales shares of the largest 5%, 1%, and 0.1% of firms change from 51.1%, 45.4%, and 37.7% to 52.6%, 42.5%, and 25.4%, respectively. See Table 6 above for empirical counterparts.

Figure 7: Optimal Firm Size-Dependent Policies, Alternative Specifications



Left: the optimal firm size-dependent marginal tax schedule $\tau_{endog}(\cdot)$, with the equilibrium multi-establishment firm cutoff and 99.99th percentile largest firm indicated, in each alternative specification. Right: changes in aggregate output in each specification from implementing each optimal policy, $\tau_{fixed}(\cdot)$, $\tau_{endog}(\cdot)$, and $\tau_{flat}(\cdot)$, including the effects of $\tau_{fixed}(\cdot)$ in the equilibrium with extensive margin responses.

6 Extensions

Our theoretical results are robust to two extensions. First, we impose uniform pricing across establishments within firms, a restriction supported by data on retail chains.[22] Second, we add richer establishment-level heterogeneity.

6.1 Uniform Pricing

Each firm now sets a single price $p(i)$ across its producing establishments. Section 5.3 computed misallocation and optimal firm size-dependent policies under this restriction. Here, we extend our characterization of distortions, Theorem 1. In the market equilibrium, let $V_{prod}(i)$ and $V_{estab}(i)$ denote the marginal social values of labor used, respectively, to expand production at firm i's open establishments, and to open and produce at its marginal establishment. In both cases, hold fixed total production labor across each firm's market equilibrium open establishments, reallocating within firms to maintain uniform pricing. We now state our result.

[22] See DellaVigna and Gentzkow (2019) for the U.S.; Daruich and Kozlowski (2023) for Argentina.

Theorem 2. *If the establishment demand elasticity $\sigma(q)$ is constant in relative output q, then the market equilibrium is efficient.*

Let $\hat{\sigma}(i)$ be the sales-weighted average demand elasticity across firm i's open establishments, equal to the production labor-weighted average. If $\sigma(q)$ is strictly decreasing for all $q > 0$, then the following are true at all firms i with positive output:

1) **Intensive margin.** *$V_{prod}(i) = W\mu(i)$, where $\mu(i) = \frac{\hat{\sigma}(i)}{\hat{\sigma}(i)-1}$ is the firm markup.*

2) **Extensive margin.** *$V_{estab}(i) \leq W\frac{\Upsilon(q_{marg}(i))}{q_{marg}(i)\Upsilon'(q_{marg}(i))}$, with equality if firm i is multi-establishment. The right-hand side is strictly increasing in $q_{marg}(i)$, and therefore so is $V_{estab}(i)$ among multi-establishment firms.*

3) **Intensive vs. extensive.** *$V_{estab}(i) < V_{prod}(j)$ at all producing firms j with $\hat{\sigma}(j) \leq \sigma(q_{marg}(i))$. By continuity, $V_{estab}(i) < V_{prod}(j)$ also holds if $\hat{\sigma}(j)$ is greater than but sufficiently close to $\sigma(q_{marg}(i))$.*

These imply an inefficient market equilibrium if there is a nondegenerate markup distribution or if a multi-establishment firm satisfies the comparison in 3).

Proof. See Appendix A.2. ■

Uniform pricing does not change the expression for extensive margin distortions, whereas intensive margin distortions are now at the firm level and depend on firm markups rather than establishment markups. There is thus a similar tension between the two margins: marginal establishments at multi-establishment firms should close in order to increase production along the intensive margin at firms whose establishments are on average weakly larger, or even somewhat smaller, with the appropriate average determined by the demand elasticity function $\sigma(\cdot)$.

6.2 Heterogeneous Productivity, Industries, and Locations

Returning to establishment-specific pricing, establishment opportunities now also vary by productivity $z(i,n) > 0$ and sector $s(i,n) \in [0,1]$, which represents an industry or location. On the supply side, establishment output is $y(i,n) = z(i,n)l(i,n)$. On the demand side, final good output is a Cobb-Douglas aggregate of sector outputs, $\ln(Y) = \int_0^1 \ln(\tilde{Y}(s))ds$. Each sector output is given by the Kimball aggregator constraint (3), but over the set of open establishments in the sector, using the same

function $\Upsilon(\cdot)$ in each sector, where an establishment's relative output is now its output over sector output, $q(i,n) = y(i,n)/\tilde{Y}(s(i,n))$. Each firm has a measure zero of establishment opportunities in each sector, so it remains infinitesimal.

Due to sector heterogeneity, different establishments can now face different degrees of competition. Specifically, inverse demand for an establishment in sector s is[23]

$$p(i,n) = \varphi(i,n)\Upsilon'(q(i,n))\tilde{D}(s)\tilde{P}(s), \tag{10}$$

where $\tilde{D}(s)$ is the sector demand index defined in (4) but computed within sector s, and $\tilde{P}(s) = Y/\tilde{Y}(s)$ is the sector price index. Despite this additional heterogeneity, our characterization of distortions along the intensive and extensive margins remains:

Corollary 1. *Theorem 1 holds in this extended environment.*

Proof. See Appendix A.1, which includes the required extension of Assumption 2. ■

Although the statement of Theorem 1 holds, an establishment's relative output no longer maps cleanly into sales or costs—due to multi-dimensional heterogeneity—but still maps one-for-one into its markup. More significantly, the marginal social value of labor along both the intensive and extensive margins will tend to be higher in industries or locations dominated by establishments with high relative output. This may affect optimal firm size-dependent policy if there is a strong relationship between a firm's size and the level of concentration in the markets where it operates.

7 Conclusion

Our findings provide takeaways for any context where firms operate multiple units with market power increasing in unit size. A high firm markup indicates that its existing units produce too little. But it does not mean the firm should operate more units. This is because firms overvalue opening new units relative to expanding larger or even somewhat smaller existing units. Optimal reallocation and policy therefore depend on which firms have an active extensive margin, the size of their marginal units, and the relative responsiveness of their intensive and extensive margins.

[23] The derivation for $\partial\tilde{Y}(s)/\partial y(i,n)$ is the same as for $\partial Y/\partial y(i,n)$ in the baseline model. Then, $p(i,n) = \partial Y/\partial y(i,n) = (Y/\tilde{Y}(s))\partial\tilde{Y}(s)/\partial y(i,n)$, where $Y/\tilde{Y}(s) = \tilde{P}(s)$.

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A Proofs

A.1 Proof of Theorem 1 and Corollary 1

We prove Theorem 1 in the more general environment of Section 6.2, thereby also proving Corollary 1. We must extend Assumption 2 so that along the ordering of firm i's establishments by operating profits over opening costs, the total demand shifter $\varphi(i,\tilde{n}(n))\tilde{D}(s(i,\tilde{n}(n)))\tilde{P}(s(i,\tilde{n}(n)))$ and productivity $z(i,\tilde{n}(n))$ are continuous in n.

Marginal social values. We develop expressions for the marginal social values, in an arbitrary allocation, of using labor to increase production at existing establishments, and to open and produce at new establishments. An allocation consists of a set of open establishments $N(i)$ at each firm, and production labor $l(i,n)$ at each establishment. Let $\tilde{N}(s)$ be the implied set of open establishments in sector s. Establishment output is implied by the production function $y(i,n) = z(i,n)l(i,n)$, each sector's output is implied by the Kimball aggregation constraint over establishments $n \in \tilde{N}(s)$, and aggregate output is implied by Cobb-Douglas aggregation across sectors.

The inverse demand curve (10) and $y(i,n) = z(i,n)l(i,n)$ imply that the marginal social value of production labor at any producing establishment in sector s is

$$\frac{\partial Y}{\partial l(i,n)} = z(i,n)\varphi(i,n)\Upsilon'(q(i,n))\tilde{D}(s)\tilde{P}(s), \tag{11}$$

where $\tilde{D}(s)$ and $\tilde{P}(s)$ are the sector demand index and price evaluated at the given allocation. Multiplying by $\frac{\tilde{Y}(s)}{z(i,n)}$ gives the derivative of aggregate output with respect

to the establishment's relative output. Then integrating over establishment relative output from 0 to $q(i,n)$ and dividing by the total labor used for the opening cost, $\kappa(i,n)$, and production, $l(i,n)$, yields the social value per unit of labor from opening the establishment and producing its output:

$$\frac{\varphi(i,n)\Upsilon(q(i,n))\tilde{Y}(s)}{\kappa(i,n)+l(i,n)}\tilde{D}(s)\tilde{P}(s). \tag{12}$$

CES demand. We now prove the first statement of the theorem: if the demand elasticity is $\sigma(q)=\sigma>1$, then the market equilibrium is efficient. To do so, we solve for the efficient allocation and show that the market equilibrium implements it. As a preliminary, observe that the definition $\sigma(q)=\frac{-\Upsilon'(q)}{q\Upsilon''(q)}$ and $\Upsilon(0)=0$ imply that $\Upsilon(q)=Bq^{\frac{\sigma-1}{\sigma}}$ for some $B>0$. Plugging into (4) for the demand index and imposing the Kimball aggregation constraint yields the sector demand index $\tilde{D}(s)=D=\frac{\sigma}{\sigma-1}$.

Given a set of open establishments, we solve for the efficient allocation of production labor. Within each sector, efficiency equalizes $\frac{\partial\tilde{Y}(s)}{\partial l(i,n)}$ across open establishments to sector productivity $\tilde{A}(s)$, which converts sector production labor $\tilde{L}_p(s)$ into sector output $\tilde{Y}_s$ via the sector production function $\tilde{Y}(s)=\tilde{A}(s)\tilde{L}_p(s)$. Using the CES $\Upsilon(\cdot)$ and D, as well as $\frac{\partial\tilde{Y}(s)}{\partial l(i,n)}=\frac{\tilde{Y}(s)}{Y}\frac{\partial Y}{\partial l(i,n)}=\frac{1}{\tilde{P}(s)}\frac{\partial Y}{\partial l(i,n)}$, equation (11) implies that

$$q(i,n)=\left(\frac{z(i,n)\varphi(i,n)B}{\tilde{A}(s)}\right)^{\sigma}\qquad \tilde{A}(s)^{\sigma-1}=B^{\sigma}\int_{(i,n)\in\tilde{N}(s)}z(i,n)^{\sigma-1}\varphi(i,n)^{\sigma}dndi, \tag{13}$$

where the second follows from plugging the first into the Kimball aggregator constraint. This is the unique efficient allocation of production labor within sectors. Across sectors, the efficient allocation equalizes $\frac{\partial Y}{\partial l(i,n)}=\frac{Y}{\tilde{Y}(s)}\tilde{A}(s)=\frac{Y}{L_p(s)}$, so production labor is constant across sectors at L_p. Thus, aggregate output is given by

$$\ln(Y)=\ln(L_p)+\int_0^1\ln(\tilde{A}(s))ds\qquad L_p=\bar{L}-\int_0^1\int_{N(i)}\kappa(i,n)dndi. \tag{14}$$

We now solve for the efficient set of open establishments. Differentiating (14) implies that opening firm i's establishment n and producing its efficient output increases aggregate output if the benefit of higher $\ln(\tilde{A}(s))$ outweighs the cost of lower $\ln(L_p)$:

$$\frac{1}{\sigma-1}\tilde{A}(s)^{1-\sigma}B^{\sigma}z(i,n)^{\sigma-1}\varphi(i,n)^{\sigma}\geq\frac{1}{L_p}\kappa(i,n). \tag{15}$$

Raising $\tilde{A}(s)^{1-\sigma}L_p$ expands the set of open establishments through (15). This lowers L_p and raises $\tilde{A}(s)$, which means a lower $\tilde{A}(s)^{1-\sigma}L_p$ because $\sigma > 1$. Thus, there is a unique fixed point, which yields the unique efficient allocation.

Finally, we show that the market equilibrium delivers this efficient allocation. All establishments have markup $\frac{\sigma}{\sigma-1} = D$, so prices are given by $p(i,n) = \frac{DW}{z(i,n)}$. Since the price is $\frac{\partial Y}{\partial y(i,n)}$, it follows from $y(i,n) = z(i,n)l(i,n)$ that $\frac{\partial Y}{\partial l(i,n)} = DW$. Hence, the market allocates production labor efficiently. Plugging the earlier characterization (13) for efficient establishment output into establishment profits yields

$$\frac{\pi(i,n)}{W} = \frac{l(i,n)}{\sigma-1} = \frac{q(i,n)\tilde{A}(s)L_p/z(i,n)}{\sigma-1} = \frac{\tilde{A}(s)^{1-\sigma}B^{\sigma}z(i,n)^{\sigma-1}\varphi(i,n)^{\sigma}L_p}{\sigma-1}.$$

Firm i opens establishment n if this expression exceeds the opening cost $\kappa(i,n)$. This is exactly condition (15), so the market implements the unique efficient allocation.

Non-CES demand. We now prove the three statements for a strictly decreasing demand elasticity $\sigma(q)$. Suppose we are in a market equilibrium. First, it follows from the definition of the markup $\mu(i,n)$ that each producing establishment's price is $p(i,n) = \frac{W\mu(i,n)}{z(i,n)}$. Since the price is $\frac{\partial Y}{\partial y(i,n)}$, it follows from $y(i,n) = z(i,n)l(i,n)$ that $\frac{\partial Y}{\partial l(i,n)} = W\mu(i,n)$. The optimal markup from (6) is $\mu(i,n) = \frac{\sigma(q(i,n))}{\sigma(q(i,n))-1}$, which is strictly increasing in $q(i,n)$ because $\sigma(q)$ is strictly decreasing. This is statement 1).

Next, by the definition of firm i's marginal establishment $n^*(i)$ in sector s, revenues $p(i,n^*(i))q_{marg}(i)\tilde{Y}(s)$ are weakly below total costs $W(\kappa(i,n^*(i)) + l(i,n^*(i)))$, where the two are equal if firm i is multi-establishment. Using the inverse demand curve (10) for the price and plugging into the denominator of (12) for the marginal social value of opening an establishment, it follows that $V_{estab}(i) \leq W\frac{\Upsilon(q_{marg}(i))}{q_{marg}(i)\Upsilon'(q_{marg}(i))}$, with equality if firm i is multi-establishment. Since $\Upsilon(\cdot)$ is twice continuously differentiable on $(0,\infty)$, the fundamental theorem of calculus implies that for all $q > 0$ and $q_0 \in (0,q)$,

$$\Upsilon(q) = \Upsilon(q_0) + \int_{q_0}^{q} \Upsilon'(\tilde{q})d\tilde{q} \qquad\qquad q\Upsilon'(q) = q_0\Upsilon'(q_0) + \int_{q_0}^{q} \left(\Upsilon'(\tilde{q}) + \tilde{q}\Upsilon''(\tilde{q})\right) d\tilde{q}.$$

Take the limit as q_0 goes to 0. $\Upsilon(q_0)$ goes to $\Upsilon(0) = 0$. Concavity of $\Upsilon(\cdot)$ means $q_0\Upsilon'(q_0) \leq \Upsilon(q_0)$ for all $q_0 > 0$. Since $q_0\Upsilon'(q_0) > 0$ and $\Upsilon(q_0)$ goes to 0, we get that

$q_0\Upsilon'(q_0)$ goes to 0. Thus, for all $q > 0$, plugging in the definition $\sigma(q) = \frac{-\Upsilon'(q)}{q\Upsilon''(q)}$ yields

$$\frac{q\Upsilon'(q)}{\Upsilon(q)} = \frac{\int_0^q \Upsilon'(\tilde{q})\frac{\sigma(\tilde{q})-1}{\sigma(\tilde{q})}d\tilde{q}}{\int_0^q \Upsilon'(\tilde{q})d\tilde{q}},$$

where the integrals are improper integrals because the integrands are undefined at 0. Thus, $\frac{q\Upsilon'(q)}{\Upsilon(q)}$ is a weighted average of $\frac{\sigma(\tilde{q})-1}{\sigma(\tilde{q})}$ for $\tilde{q} \in (0, q)$, with strictly positive weights, $\Upsilon'(\tilde{q})$. Since $\sigma(\cdot)$ is strictly decreasing, it follows that the reciprocal, $\frac{\Upsilon(q)}{q\Upsilon'(q)}$, is strictly increasing in q, completing the proof of statement 2). Moreover, it follows that $\frac{\Upsilon(q(i,n))}{q(i,n)\Upsilon'(q(i,n))} < \frac{\sigma(q(i,n))}{\sigma(q(i,n))-1} = \mu(i,n)$. Using statement 1) then yields statement 3).

Finally, a nondegenerate distribution of establishment relative output implies markup variation, so by 1), we can raise output by reallocating production labor. Otherwise, if there are multi-establishment firms, then by 3), we can raise output by closing marginal establishments and producing more at remaining establishments.

A.2 Proof of Theorem 2

There is no sector or productivity heterogeneity, so $y(i,n) = l(i,n)$.

CES demand. We begin with the first statement: if the demand elasticity is $\sigma(q) = \sigma > 1$, then the market equilibrium is efficient. In the market equilibrium, uniform pricing does not bind because all firms optimally set the same price $\frac{\sigma}{\sigma-1}W$ at all open establishments. In the planner problem, uniform pricing does not bind because the planner optimally equalizes $\frac{\partial Y}{\partial l(i,n)}$ across open establishments, and $\frac{\partial Y}{\partial l(i,n)} = \frac{\partial Y}{\partial y(i,n)}$. Thus, the market equilibrium and the unique efficient allocation are identical with uniform vs. establishment-specific pricing. The result then follows from Theorem 1.

Non-CES demand. For the rest of the proof, suppose the demand elasticity $\sigma(q)$ is strictly decreasing in q. We first show statement 1). At a producing establishment at firm i, the marginal social value of production labor, $\frac{\partial Y}{\partial l(i,n)}$, is the firm price because demand implies that $p(i) = \frac{\partial Y}{\partial y(i,n)}$. Moreover, $\frac{\partial Y}{\partial l(i,n)} \leq p(i)$ at any non-producing open establishments. Thus, production labor is efficiently allocated within firms. It follows that the marginal social value of increasing firm i's production labor is $V_{prod}(i) = p(i)$; any implied within-firm reallocation of another firm's production labor to maintain uniform pricing has no effect on aggregate output by the envelope theorem. Plugging in the definition of the markup as price over marginal cost, we get $V_{prod}(i) = W\mu(i)$.

We now solve for the markup. Firm i's first order condition for its price $p(i)$ is

$$\int_{N(i)} \left(y(i,n) + (p(i) - W) \frac{\partial y(i,n)}{\partial p(i)} \right) dn = 0,$$

where the first term is the direct increase in revenue from a higher price, and the second term is the fall in profits from the resulting fall in output sold. Plugging in $\sigma(q(i,n)) = \frac{-p(i)}{y(i,n)} \frac{\partial y(i,n)}{\partial p(i)}$, $y(i,n) = l(i,n)$, and $\mu(i) = \frac{p(i)}{W}$ yields

$$\int_{N(i)} l(i,n) \left[1 - \left(1 - \mu(i)^{-1} \right) \sigma(q(i,n)) \right] dn = 0.$$

Thus, $\mu(i) = \frac{\hat{\sigma}(i)}{\hat{\sigma}(i)-1}$, where $\hat{\sigma}(i) > 1$ is the production labor-weighted average demand elasticity across firm i's establishments, which is also the sales-weighted average because sales over production labor is constant at $p(i)$. This completes statement 1).

On the extensive margin, relative output $q_{marg}(i)$ at firm i's marginal establishment is implied by firm i's price. The social value of opening and producing at this establishment is the same function of $q_{marg}(i)$ and $\kappa(i, n^*(i))$ as with establishment-specific pricing because, as with the intensive margin, any implied within-firm reallocation of production labor to maintain uniform pricing has no effect on aggregate output by the envelope theorem. The same is true for the private value because firm i chooses whether to open an establishment based only on that establishment's profits, taking its price $p(i)$ as given by the envelope theorem. Thus, statement 2) follows from Theorem 1. It also follows from Theorem 1 that if $\sigma(q_{marg}(i)) \geq \hat{\sigma}(j) > 1$, then $V_{estab}(i) < W \frac{\sigma(q_{marg}(i))}{\sigma(q_{marg}(i))-1} \leq W \frac{\hat{\sigma}(j)}{\hat{\sigma}(j)-1}$, which, using statement 1), implies statement 3).

Finally, by 1), markup variation means we can raise output by reallocating production labor. If a multi-establishment firm satisfies the comparison in 3), we can raise output by closing its marginal establishment to raise firm j's production labor.